

Mathematical Model of a Pregnancy Test

Aneesh Chatterjee (Reg. No. : MATS632)
Sri Venkateswara College, University of Delhi

Supervisors: Profs. Amiya Kumar Pani, P. Danumjaya, Anil Kumar

Internship done in the Department of Mathematics at
BITS Pilani K.K Birla Goa Campus, Zuarinagar, Goa - 403726

June 10th 2023 - August 5th 2023

1 Introduction

Let us begin with an overview of the phenomena being modelled. In this case, a simple pregnancy test. The test is administered using a dipstick with an antigen on its surface dipped into a urine sample collected from the person being tested, stored in a vessel. Our goal is to develop a mathematical model for the above phenomenon and then study it. This enables us to gain a better understanding of the behaviour of this model over time, while learning various new concepts related to partial differential equations and mathematical modelling along the way.

2 Forming The Model

This section deals with developing the mathematical model describing the pregnancy test. First let us understand the chemical reaction at play here. In this test, an antigen compound reacts with an antibody compound in order to produce a product that is blue in colour.

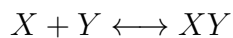
2.1 Assumptions:

Assume that

- One molecule of X combines with one molecule of Y to give one molecule of XY
- The concentration of the antibody compound in the urine is initially the same all throughout the vessel

2.2 Chemical reaction

The chemical reaction is as follows



The double arrow simply indicates that forward and backward reactions are both taking place.

2.3 Defining important terms

Now let us define some terms that will be used for developing the model.

X : Antigen compound

Y : Antibody compound

XY : Blue product compound

$[X]$: Concentration of compound X

$[Y]$: Concentration of compound Y

$[XY]$: Concentration of compound XY

k_1 : Forward reaction rate

k_{-1} : Backward reaction rate

a_0 : initial concentration of compound X

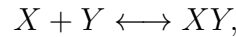
c_0 : initial concentration of compound Y at sidewall

d : distance between the vessel edge and surface (vessel assumed to be $2d$)

D : diffusion coefficient

2.4 Modelling

Let us consider d as the distance from the edge of the vessel to the surface of the dip-stick in the $x - axis$. When a dip-stick with antigen on its surface is dipped into the urine collected in the vessel, if the woman is pregnant antibodies are present in the urine, and these antibodies react with the antigen to produce a blue colour. Using the symbols as above, we have the reaction:



where we denote the forward and backward reaction rates by k_1 and k_{-1} respectively. Now since $[X]$ denotes the antigen concentration, by conservation of mass

$$\frac{\partial[X]}{\partial t} + \frac{\partial q}{\partial x} = 0, \quad a < x < d, \quad t > 0,$$

where q is the flux. Now using Fick's Law

$$q \propto \frac{\partial[X]}{\partial x}.$$

Hence,

$$q = -D \frac{\partial[X]}{\partial x},$$

where D is the diffusivity. Thus, we obtain

$$\frac{\partial[X]}{\partial t} = D \frac{\partial^2[X]}{\partial x^2}, \quad 0 < x < d, \quad t > 0,$$

where x is distance from the base of the vessel and t is time.

Now we need to define some initial and boundary conditions so that we can close the system.

Firstly, we know that the concentration of X does not change at $x = 0$, i.e, at the rim of the vessel. This gives us the condition:

$$\left. \frac{\partial[X]}{\partial x} \right|_{x=0} = 0.$$

Now, at time $t = 0$, the concentration of X will be equal to its initial concentration. Hence,

$$[X]_{t=0} = a_0.$$

Now, we need to determine a boundary condition at $x = d$. But first, we introduce a change in notation for more clarity.

$u(x, t)$ = concentration of X (i.e $[X]$)

and

$\theta(t)$ = concentration of XY (i.e $[XY]$)

Since the change in concentration of X depends on the change in concentration of Y and XY , by the Law of Mass Action, it follows that

$$D \frac{\partial u}{\partial x}(x, t) \Big|_{x=d} = k_{-1}\theta(t) - k_1 u(d, t)[Y]$$

Now the initial concentration of Y at the sidewall is c_0 . This c_0 is depleted by the amount of X used in the reaction, since equal

amounts of X and Y combine to give XY . Therefore,

$$[Y](t) = c_0 - (a_0 d - \int_0^d u(x, t) dx).$$

Substituting this in the previous equation we obtain

$$D \frac{\partial u}{\partial x}(x, t) \Big|_{x=d} = k_{-1}\theta(t) - k_1 u(d, t) \{c_0 - (a_0 d - \int_0^d u(x, t) dx)\}.$$

Due to the principle of Conservation of Mass, the total amount of X must be conserved, either present in the solution or in the form of compound XY . Therefore,

$$\int_0^d u(x, t) dx + \theta(t) = a_0 d.$$

Replacing this value, we arrive at

$$D \frac{\partial u}{\partial x}(x, t) \Big|_{x=d} = k_{-1}\theta(t) - k_1 u(d, t)[c_0 - \theta(t)].$$

Finally, summarising all the previous equations, we are left with the following IBVP:

$$\frac{\partial u}{\partial t}(x, t) = D \frac{\partial^2 u}{\partial x^2}(x, t), \quad 0 < x < d, \quad (1)$$

$$\frac{\partial u}{\partial x}(x, t) \Big|_{x=0} = 0, \quad t > 0, \quad (2)$$

$$D \frac{\partial u}{\partial x}(x, t) \Big|_{x=d} = k_{-1}\theta(t) - k_1 u(d, t)[c_0 - \theta(t)], \quad t > 0, \quad (3)$$

$$u(x, 0) = a_0, \quad 0 < x < d. \quad (4)$$

3 Non-Dimensionalisation

The IBVP we have described above contains a lot of constants, making the equations hard to deal with when solving. In order to take care of this problem, we now non dimensionalise the independent variables and scale the dependent variables by some constants, that is,

$$x' = \frac{x}{d}, \quad t' = \frac{D}{d^2} t.$$

Also scaling

$$u'(x', t') = \frac{u(x, t)}{a_0}, \quad \theta'(t') = \frac{\theta(t)}{c_0}.$$

Now we must compute the various derivatives, using chain rule:

$$\begin{aligned} \frac{\partial}{\partial x} &= \frac{\partial x'}{\partial x} \cdot \frac{\partial}{\partial x'} = \frac{1}{d} \cdot \frac{\partial}{\partial x'}, \\ \frac{\partial}{\partial t} &= \frac{\partial t'}{\partial t} \cdot \frac{\partial}{\partial t'} = \frac{D}{d^2} \cdot \frac{\partial}{\partial t'}. \end{aligned}$$

Replacing these values in equation (1) we obtain

$$\frac{a_0 D}{d^2} \cdot \frac{\partial u'}{\partial t'}(x', t') = \frac{D a_0}{d^2} \cdot \frac{\partial^2 u'}{\partial x'^2}(x', t').$$

Hence,

$$\frac{\partial u'}{\partial t'}(x', t') = \frac{\partial^2 u'}{\partial x'^2}(x', t').$$

Our initial condition at $t = 0$ becomes

$$u'(x, 0) = 1.$$

The boundary condition at $x = 0$ becomes

$$\left. \frac{\partial u'}{\partial x'} \right|_{x'=0} = 0.$$

Now, we need to deal with our other boundary condition at $x = d$.

Substituting the values, we arrive at

$$\frac{D a_0}{d} \cdot \left. \frac{\partial u'}{\partial x'} \right|_{x'=1} = k_{-1} c_0 \theta'(t') - k_1 a_0 u'(1, t') (c_0 - c_0 \theta(t)),$$

and,

$$\left. \frac{\partial u'}{\partial x'} \right|_{x'=1} = \frac{d c_0}{D a_0} [k_{-1} \theta'(t') - k_1 a_0 u'(1, t') (1 - \theta(t))].$$

Therefore,

$$\left. \frac{\partial u'}{\partial x'} \right|_{x'=1} = \frac{k_1 d c_0}{D} \left[\frac{k_{-1}}{k_1 a_0} \theta'(t') - u'(1, t')(1 - \theta(t)) \right].$$

For convenience, we introduce the following terms:

Molar ratio $m = \frac{c_0}{a_0 d}$,

Reaction time scale ratio $L = \frac{k_{-1}}{k_1 a_0}$,

Diffusion reaction time scale $E = \frac{(k_1 a_0 + k_{-1}) d^2}{D}$,

Now there holds

$$\frac{Em}{1+L} = \frac{(k_1 a_0 + k_{-1}) d^2 c_0}{\left(1 + \frac{k_{-1}}{k_1 a_0}\right) a_0 d^2 D},$$

that is,

$$\frac{Em}{1+L} = \frac{k_1 d c_0}{D}.$$

Therefore, the boundary condition reduces to

$$\left. \frac{\partial u'}{\partial x'} \right|_{x'=1} = \frac{Em}{1+L} [L \theta'(t') - u'(1, t')(1 - \theta(t))].$$

We shall also be using the following equation:

$$\theta(t) + \int_0^d u(x, t) dx = a_0 d.$$

From this, we arrive at

$$c_0 \theta'(t') + a_0 d \int_0^1 u'(x', t') dx' = a_0 d,$$

and hence,

$$m \theta'(t') + \int_0^1 u'(x', t') dx' = 1.$$

Now, dropping the prime notation, we are left with the following set of simplified equations

$$\frac{\partial u}{\partial t}(x, t) = \frac{\partial^2 u}{\partial x^2}(x, t), \quad 0 < x < 1, \quad t > 0, \quad (5)$$

$$u(x, 0) = 1, \quad 0 < x < 1, \quad (6)$$

$$\left. \frac{\partial u}{\partial x} \right|_{x=0} = 0, \quad t > 0, \quad (7)$$

$$\left. \frac{\partial u}{\partial x} \right|_{x=1} = \frac{Em}{1+L} [L\theta(t) - u(1, t)(1 - \theta(t))] \quad t > 0, \quad (8)$$

$$m\theta(t) + \int_0^1 u(x, t) dx = 1, \quad t > 0. \quad (9)$$

4 Reduction to Integro-Differential Equation

Even after non-dimensionalisation, we may not be able to solve the IBVP analytically. Therefore, we shall now use convolution and Laplace Transforms to reduce it into a nonlinear Volterra Integro-Differential Equation.

First, we differentiate (9) with respect to t to obtain

$$m \frac{d\theta}{dt}(t) + \int_0^1 \frac{\partial u}{\partial t}(x, t) dx = 0. \quad (10)$$

From (5)

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

And on using (10), we arrive at

$$m \frac{d\theta}{dt}(t) + \int_0^1 \frac{\partial^2 u}{\partial x^2}(x, t) dx = 0.$$

On simplification

$$-m \frac{d\theta}{dt}(t) = \frac{\partial u}{\partial x}(1, t). \quad (11)$$

Now, we denote the Laplace transform of u by $\bar{u}$, where

$$\bar{u}(x, p) = \int_0^\infty u(x, t)e^{-pt} dt, \quad p > 0.$$

Taking the Laplace Transform of (5) and using (6) we find that

$$\frac{\partial^2 \bar{u}}{\partial x^2}(x, p) = p\bar{u}(x, p) - 1.$$

On solving, the general solution is given by

$$\bar{u}(x, p) = \frac{1}{p} + A(p)\cosh(\sqrt{p}x) + B(p)\sinh(\sqrt{p}x), \quad (12)$$

where $A(p)$ and $B(p)$ are arbitrary constants.

Now, we differentiate with respect to x to obtain

$$\frac{\partial \bar{u}}{\partial x}(x, p) = A(p)\sqrt{p}\sinh(\sqrt{p}x) + B(p)\sqrt{p}\cosh(\sqrt{p}x).$$

Since $\frac{\partial u}{\partial x}(0, t) = 0$, $B(p) = 0$, and therefore,

$$\frac{\partial \bar{u}}{\partial x}(x, p) = A(p)\sqrt{p}\sinh(\sqrt{p}x).$$

On solving,

$$A(p) = \frac{\frac{\partial \bar{u}}{\partial x}(x, p)}{\sqrt{p}\sinh(\sqrt{p}x)}.$$

Substituting this into the expression for $\bar{u}(x, p)$, we arrive at

$$\bar{u}(x, p) = \frac{1}{p} + \frac{1}{\sqrt{p}} \cdot \frac{\partial \bar{u}}{\partial x}(x, p) \cdot \cotanh(\sqrt{p}x).$$

Now at $x = 1$

$$\bar{u}(1, p) = \frac{1}{p} + \bar{u}_x(1, p)\bar{k}(p),$$

where

$$\bar{u}_x(1, p) = \frac{\partial \bar{u}}{\partial x}(1, p),$$

and

$$\bar{k}(p) = \frac{\cotanh(\sqrt{p})}{\sqrt{p}}.$$

Thus using the inverse Laplace transform, we know that $\mathcal{L}^{-1}\left(\frac{1}{p}\right) = 1$, and furthermore, by the convolution theorem we obtain,

$$u(1, t) = 1 + \int_0^t \frac{\partial u}{\partial x}(1, s)k(t-s)ds,$$

where

$$k(t) = \mathcal{L}^{-1}(\bar{k}(p)) = (\pi t)^{-1/2} \left(1 + 2 \sum_{n=1}^{\infty} \exp\left(\frac{-n^2}{t}\right) \right).$$

Now using (11)

$$u(1, t) = 1 - m \int_0^t \frac{d\theta}{ds}(s)k(t-s)ds.$$

From (8), (11) and (12)

$$-m \frac{d\theta}{dt} = \frac{Em}{1+L} \left[L\theta(t) - (1-\theta(t)) \left\{ 1 - m \int_0^t \frac{d\theta}{ds}(s)k(t-s)ds \right\} \right],$$

and hence, a non-linear Volterra Integro-Differential equation in θ as

$$-\frac{d\theta}{dt} = \frac{E}{1+L} \left[L\theta(t) - (1-\theta(t)) \left\{ 1 - m \int_0^t \frac{d\theta}{ds}(s)k(t-s)ds \right\} \right].$$

5 Numerical Method

Our Integro-Differential Equation is of the form

$$-\frac{d\theta}{dt} = \frac{E}{1+L} \left[L\theta(t) - (1-\theta(t)) \left\{ 1 - m \int_0^t \frac{d\theta}{ds}(s)k(t-s)ds \right\} \right]$$

and hence,

$$-\frac{d\theta}{dt} = E\theta(t) - \frac{E}{1+L} + \frac{Em}{1+L}(1-\theta(t)) \int_0^t \frac{d\theta}{ds}k(t-s)ds$$

Setting $\frac{E}{1+L} = C$, we now arrive at

$$\theta'(t) = C - E\theta(t) + Cm(1-\theta(t)) \int_0^t \frac{d\theta}{ds}k(t-s)ds$$

Therefore, our set of equations is

$$\theta'(t) = C - E\theta(t) + Cm(1-\theta(t)) \int_0^t \theta'(s)k(t-s)ds, \quad (13)$$

$$u(1, t) = 1 - m \int_0^t k(t-s)\theta'(s)ds, \quad (14)$$

$$\text{with } k(t) = (\pi t)^{-1/2} \left(1 + 2 \sum_{n=1}^{\infty} \exp\left(\frac{-n^2}{t}\right) \right). \quad (15)$$

First, we need to truncate the infinite series in $k(t)$, suppose to l terms.

$$k_l(t) = (\pi t)^{-1/2} \left(1 + 2 \sum_{n=1}^l \exp\left(\frac{-n^2}{t}\right) \right), \quad t \in [0, T]$$

Given T , we select $\xi > 0$ such that

$$|k(t) - k_l(t)| < \xi \quad \forall t \in [0, T].$$

Now,

$$\left| k(t) - k_l(t) \right| = \left| (\pi t)^{-1/2} \left(1 + 2 \sum_{n=1}^{\infty} \exp\left(\frac{-n^2}{t}\right) \right) - (\pi t)^{-1/2} \left(1 + 2 \sum_{n=1}^l \exp\left(\frac{-n^2}{t}\right) \right) \right|$$

and hence,

$$\left| k(t) - k_l(t) \right| = \left| 2(\pi t)^{-1/2} \left(1 + 2 \sum_{n=l}^{\infty} \exp\left(\frac{-n^2}{t}\right) \right) \right|$$

Now

$$\left| k(t) - k_l(t) \right| < \left(\frac{2}{\pi} \right)^{1/2} \int_{(2/t^3)^{1/2}l}^{\infty} \exp\left(\frac{x^2}{2}\right) dx.$$

Now, the Cumulative Distribution Function (CDF) for a Standard Normal Distribution is given by

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp\left(\frac{x^2}{2}\right) dx.$$

Therefore,

$$\phi((2/t)^{1/2}l) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{(2/t)^{1/2}l} \exp\left(\frac{x^2}{2}\right) dx,$$

and Since $\phi(z) = 1 - \phi(z)$.

$$1 - \phi((2/t)^{1/2}l) = \left(\frac{2}{\pi} \right)^{1/2} \int_{(2/t)^{1/2}l}^{\infty} \exp\left(\frac{x^2}{2}\right) dx$$

Therefore,

$$\begin{aligned} & \left| k(t) - k_l(t) \right| < 1 - \phi((2/t)^{1/2}l) \\ \Rightarrow & \left| k(t) - k_l(t) \right| < \xi \text{ if } \phi((2/t)^{1/2}l) < 1 - \xi \end{aligned}$$

Now, we use a product integration method and approximation

$$\int_0^t f(s)g(s)ds = \sum_{j=0}^{N-1} \int_{t_j}^{t_{j+1}} f(s)g(s)ds \approx \sum_{j=0}^{N-1} \int_{t_j}^{t_{j+1}} p_n(s)g(s)ds$$

Here, $p_n(s)$ is a Lagrange Polynomial used to approximate $f(s)$, and

we have divided the interval $[0, T]$ into N uniform sub-intervals

$$[t_j, t_{j+1}], \quad j = 0, 1, 2, \dots, (N-1) \text{ with } t_{j+1} - t_j = \Delta t.$$

Now, in (13), replacing $k(t)$ by $k_l(t)$ and using the Product Integration Method

$$\theta'(t_i) = C - E\theta(t) - Cm(1 - \theta(t)) \sum_{j=0}^{i-1} \int_{t_j}^{t_{j+1}} k_l(t-s)\theta'(s)ds. \quad (16)$$

First, for $t \in [t_j, t_{j+1}]$, we approximate $\theta(t)$ by the Lagrange Polynomial

$$\begin{aligned} \theta(t) &= \frac{t - t_{j+1}}{t_j - t_{j+1}} \cdot \theta(t_j) + \frac{t - t_j}{t_{j+1} - t_j} \cdot \theta(t_{j+1}) \\ &= \frac{1}{\Delta t} [(t - t_j)\theta(t_{j+1}) - (t - t_{j+1})\theta(t_j)]. \end{aligned}$$

We also have the approximation

$$\int_{t_j}^{t_{j+1}} \exp\left(\frac{-n^2}{t_i - s}\right) \cdot \frac{1}{(t_i - s)^{1/2}} ds \approx \exp\left(\frac{-n^2}{t_i - s}\right) \int_{t_j}^{t_{j+1}} \frac{1}{(t_i - s)^{1/2}} ds.$$

Using this, and replacing the derivative by difference, we obtain the following scheme

$$\theta_0 = 0$$

$$\frac{\theta_i - \theta_{i-1}}{\Delta t} = C - E\theta_i - Cm(\pi)^{-1/2}(1 - \theta_i)\Delta t \sum_{j=0}^{i-1} \gamma(i-j) \frac{\theta_{j+1} - \theta_j}{\Delta t}, \quad (17)$$

where

$$\gamma(i-j) = \frac{1}{\Delta t} \left[1 + 2 \sum_{n=1}^l \exp\left(\frac{-n^2}{t_i - t_j}\right) \right] \int_{t_j}^{t_{j+1}} \frac{ds}{(t_i - s)^{1/2}}.$$

Substituting the value of $\gamma(i - j)$ into (16) and multiplying by Δt , we arrive at the following expression

$$\theta_i = \theta_{i-1} + C\Delta t - E\theta_i\Delta t - Cm(\pi)^{-1/2}(1 - \theta_i) \sum_{j=0}^{i-1} h_j(\theta_{j+1} - \theta_j)$$

where

$$h_j = \left[1 + 2 \sum_{n=1}^l \exp\left(\frac{-n^2}{t_i - t_j}\right) \right] \int_{t_j}^{t_{j+1}} \frac{ds}{(t_i - s)^{1/2}}.$$

Therefore, we can separate the expression for θ_i as follows

$$\theta_i = g_{i-1} + F(\theta_i)$$

where

$$g_{i-1} = \theta_{i-1} + C\Delta t - Cm(\pi)^{-1/2} \left(\sum_{j=0}^{i-2} h_j(\theta_{j+1} - \theta_j) - h_{i-1}\theta_{i-1} \right) \quad (18)$$

and

$$F(\theta_i) = -E\theta_i\Delta t - Cm(\pi)^{-1/2} \left(h_{i-1}\theta_i - \theta_i \sum_{j=0}^{i-1} h_j(\theta_{j+1} - \theta_j) \right). \quad (19)$$

5.1 Implementing the Numerical Method

First consider the expression

$$\theta_i = g_{i-1} + F(\theta_i).$$

We now solve this iteratively using the following scheme at each time level $t = t_i$

$$\theta_0 = 0$$

$$\theta_i^{(0)} = \theta_{i-1}$$

$$\theta_i^{(N)} = g_{i-1} + F(\theta_i^{(N-1)}), \quad n = 1, 2, 3 \dots$$

Here, our initial guess for each value of θ_i is the previous value θ_{i-1} , using this initial guess as a starting point we iterate on θ , in practice this iteration only needs to be done 2 or 3 times. The code used performs the iteration 3 times for each i .

Also, having found θ_i , calculating $u(1, t_i)$ is fairly straightforward. The approximation for u_i is given by the following expression

$$u_0 = 1$$

$$u_i = 1 - m\pi^{-1/2} \sum_{j=0}^{i-1} \gamma(i-j)(\theta_{j+1} - \theta_j).$$

The following code was used to implement it.

```

1 function [lsum] = expsum(l,a,b)
2 % Sum of exponentials inside kernel
3 % Calculates specifically the exponential sum inside the kernel
4 lsum=0;
5 for i = 1:l
6     lsum = lsum + exp((-i).^2)./(a-b));
7 end
8
9 end

```

Figure 1: expsum function

Figure 1 is a user defined function used to calculate the quantity

$$\sum_{n=1}^l \exp\left(\frac{-n^2}{t_i - t_j}\right)$$

```

1 function [kval] = kernel(i,j,l,step)
2 %This function calculates the value of the kernel without taking summation
3 %fun1 = @(s,t) (t-s).^(-1/2);
4 fun1 = @(s,t) (t -s).^(-1/2);
5 ti = i.*step;
6 tj = j.*step;
7 kval = (1 + 2.*expsum(l,ti,tj)).*integral(@(s)fun1(s,ti),tj,(tj + step));
8 end

```

Figure 2: kernel function

Figure 2 is a user defined function to calculate the quantity

$$h_j = \left[1 + 2 \sum_{n=1}^l \exp \left(\frac{-n^2}{t_i - t_j} \right) \right] \left[\int_{t_j}^{t_{j+1}} \frac{ds}{(t_i - s)^{1/2}} \right]$$

```

1 function [gval] = gil(i,step,C,m,l,final)
2 %This function calculates g_(i-1) for each i value. Put in a function for
3 %simplicity
4 k_sum = 0;
5 for j = 1:i-1
6     k_sum = k_sum + kernel(i,j,l,step).*(final(j+1) - final(j));
7 end
8 gval = final(j+1) + C.*step - C.*m.*(pi).^(-1/2).*k_sum + C.*m.*(pi).^(-1/2).*kernel(i,j,l,step)*final(i);
9 end

```

Figure 3: gil function (used to calculate g_{i-1})

```

1 %MAIN CODE FOR PICARD ITERATION
2 clc, clearvars, close all
3
4 %Defining initial variables
5 T = 0.5;
6 N = 100;
7 l = 100;
8 n = 3;
9 step = T/N;
10 theta = 0;
11 uit = 0;
12 final_theta = zeros(1,N+1);
13 final_u = ones(1,N+1);
14 C = 99.0099; E = 100; m = 0.1; Fsum = 0;
15 for i = 2:N
16     gval = g11(i,step,C,m,l,final_theta);
17     for k = 1:n
18         for j1 = 1:i-1
19             v = kernel(i,j1,l,step);
20             Fsum = Fsum + v.*(final_theta(j1+2)-final_theta(j1+1));
21         end
22         ksum2 = kernel(i,j1,l,step);
23         %theta
24         theta = gval + (-E.*theta.*step - C.*m.*(pi).^(-1/2).*ksum2.*final_theta(i+1) - final_theta(i+1).*Fsum);
25         uit = 1 + (m*(pi)^(-1/2)*Fsum)/step;
26         %theta
27         Fsum = 0;
28     end
29     final_theta(i) = theta;
30     final_u(i) = uit;
31 end
32 disp(final_theta)
33 disp(final_u)

```

Figure 4: Main code Part 1

```

34
35 for p = 1:length(final_theta)
36     ti = (p-1)*step;
37     theta_i = final_theta(p);
38     u_i = final_u(p);
39     a1 = plot(ti,theta_i,'r.','LineWidth',5);
40     M1 = "\theta(t)";
41     xlabel('t')
42     hold on
43     a2 = plot(ti,u_i,'b.','MarkerSize',5);
44     M2 = "u(1,t)";
45     legend([a1,a2],[M1,M2])
46 end

```

Figure 5: Main code Part 2

6 Observations

The following graphs show the behaviour of the system for different parameter values.

In figure 6, we have chosen parameters $C = 99.099$, $E = 100$, $m = 0.1$. Here, the reaction time scale ratio $L = 0.01$. In this

case, since $E \gg 0$, we are dealing with a diffusion limited system. Further, the molar ratio $m = 0.1$. Therefore, the ratio of initial concentrations favours the compound X . Due to this, the concentration $u(1, t)$ decreases quite rapidly and reaches equilibrium around 0.6. Equal amounts of compound X and Y are consumed in the reaction, therefore, the amount of compound Y limits the reaction. Meanwhile, the amount of compound X used during the reaction is relatively low. The concentration of the product $\theta(t)$ increases rapidly and stabilises around 0.7.

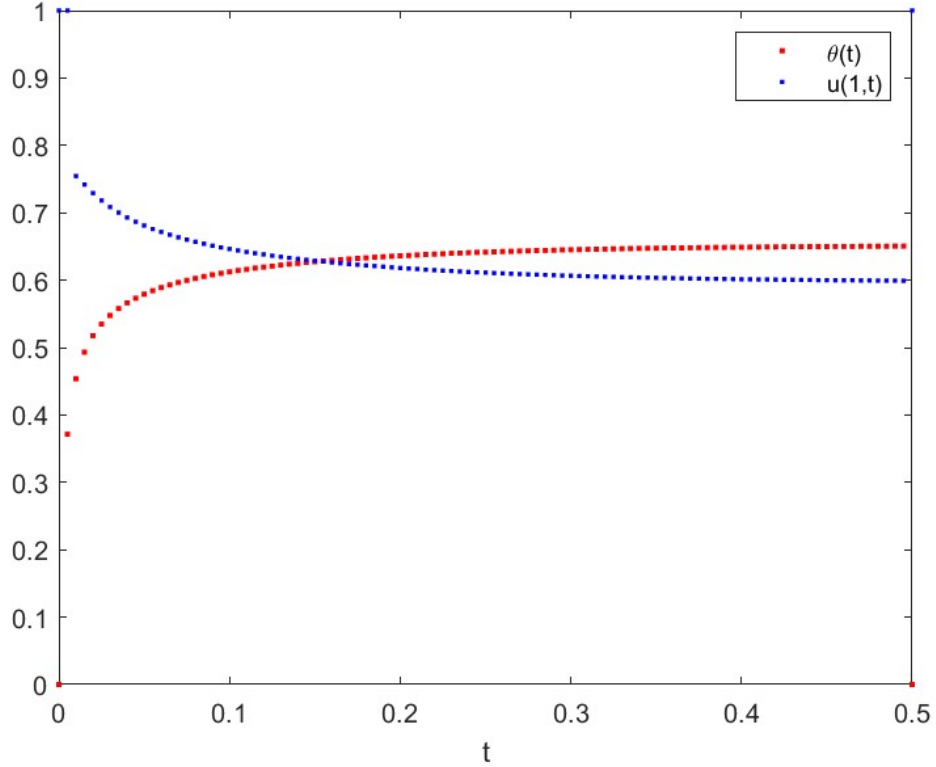


Figure 6: $C = 99.099$, $E = 100$, $m = 0.1$

In figure 7, we choose parameters $C = 9.99$, $E = 10$, $m = 0.925$. The value for E is lower than in figure 6, and the reaction time scale ratio $L = 0.001$, which is also less than in figure 6. Here, the concentration $\theta(t)$ rises quickly and stabilises around 0.2, while $u(1, t)$ drops to around 0.05. Both these values are lower than those

observed in figure 6. This is because the molar ratio $m = 0.925$ is set higher, meaning the ratio of initial concentrations favours the compound Y . Due to this, the concentration of compound X is limiting the reaction, and the amount of compound X used is relatively high.

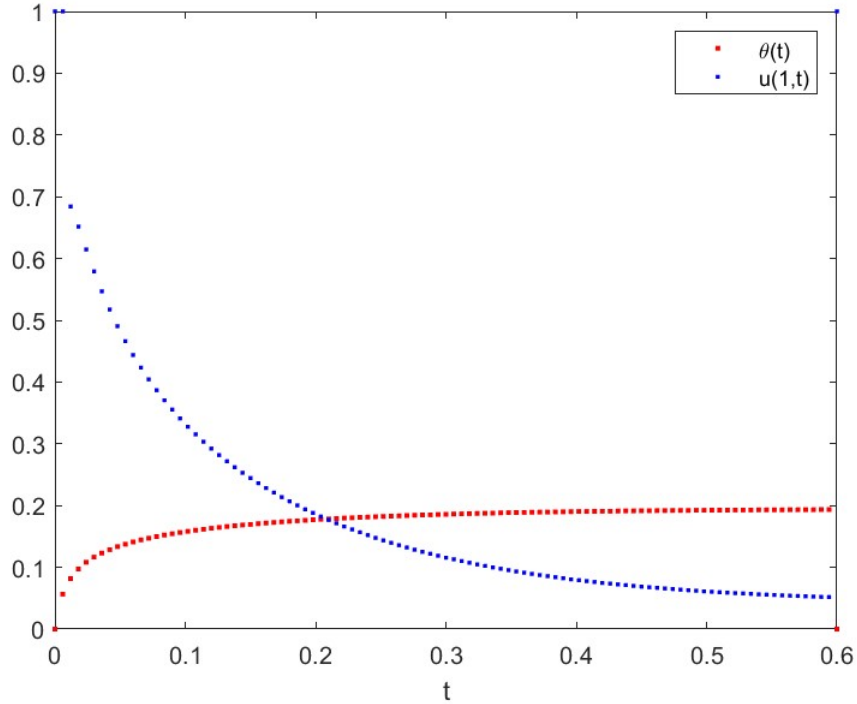


Figure 7: $C = 9.99$, $E = 10$, $m = 0.925$

7 Conclusion

In summary, we have analysed a model for a simple pregnancy test, described by the diffusion equation coupled with a nonlinear boundary condition. Then, in order to solve the system more easily we have non-dimensionalised all the equations, and used convolution and the Laplace Transform to transform them into a numerically solvable form, i.e, a Non-Linear Volterra Integro Differential Equation. Subsequently, we have developed a numerical method

to solve the equation using some approximations, and implemented the method using Matlab. In the end, we plotted two graphs representing the system for different parameter values and studied its behaviour under these conditions.

8 Acknowledgement

I would like to thank the Academy for its financial support through SRFP. More over, I would like to acknowledge the infrastructural support and help provided by the BITS-Pilani K.K. Birla Goa Campus, and most importantly, my mentors for providing guidance and wisdom throughout the length of my internship.

9 References

1. Msc. in Industrial Mathematics, *Heat and Mass Transfer*, University of Strathclyde Department of Mathematics.
2. Evans, L.C. (2010). *Partial Differential Equations*. United States: American Mathematical Society.
3. Corduneanu, C. (2008). *Integral Equations and Applications*. United Kingdom: Cambridge University Press.